

KNOWLEDGE AND REASONING

Propositional Inference



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Outline

- Knowledge-based agents
- The Wumpus world
- Inference with Propositional logic
- Propositional theorem proving
- Effective propositional model checking

Knowledge-based agents

Problem-solving agents

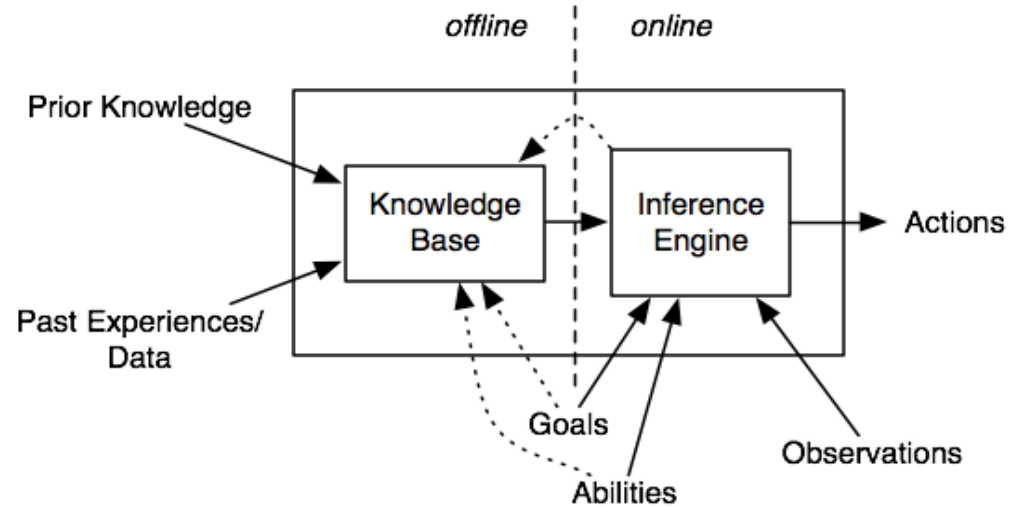
- These agents **know things in a very limited, inflexible sense**.
 - They know what actions are available and the results of these actions, but they do not know general facts.
 - E.g., an 8-puzzle agent doesn't know that two tiles cannot occupy the same space.
- The **factored representation in CSP** enables some parts of the agent to **work domain-independently**.

Knowledge-based (KB) agents

- **Knowledge-based agents** perform **reasoning** over **an internal representation** of knowledge to **decide what actions to take**.
- Supported by **logic** – a general class of representation
- **Combine and recombine information** to suit myriad purposes
 - Accept new tasks in the form of explicitly described goals
 - Achieve competence by learning new knowledge of the environment
 - Adapt to changes by updating the relevant knowledge

Knowledge-based agents

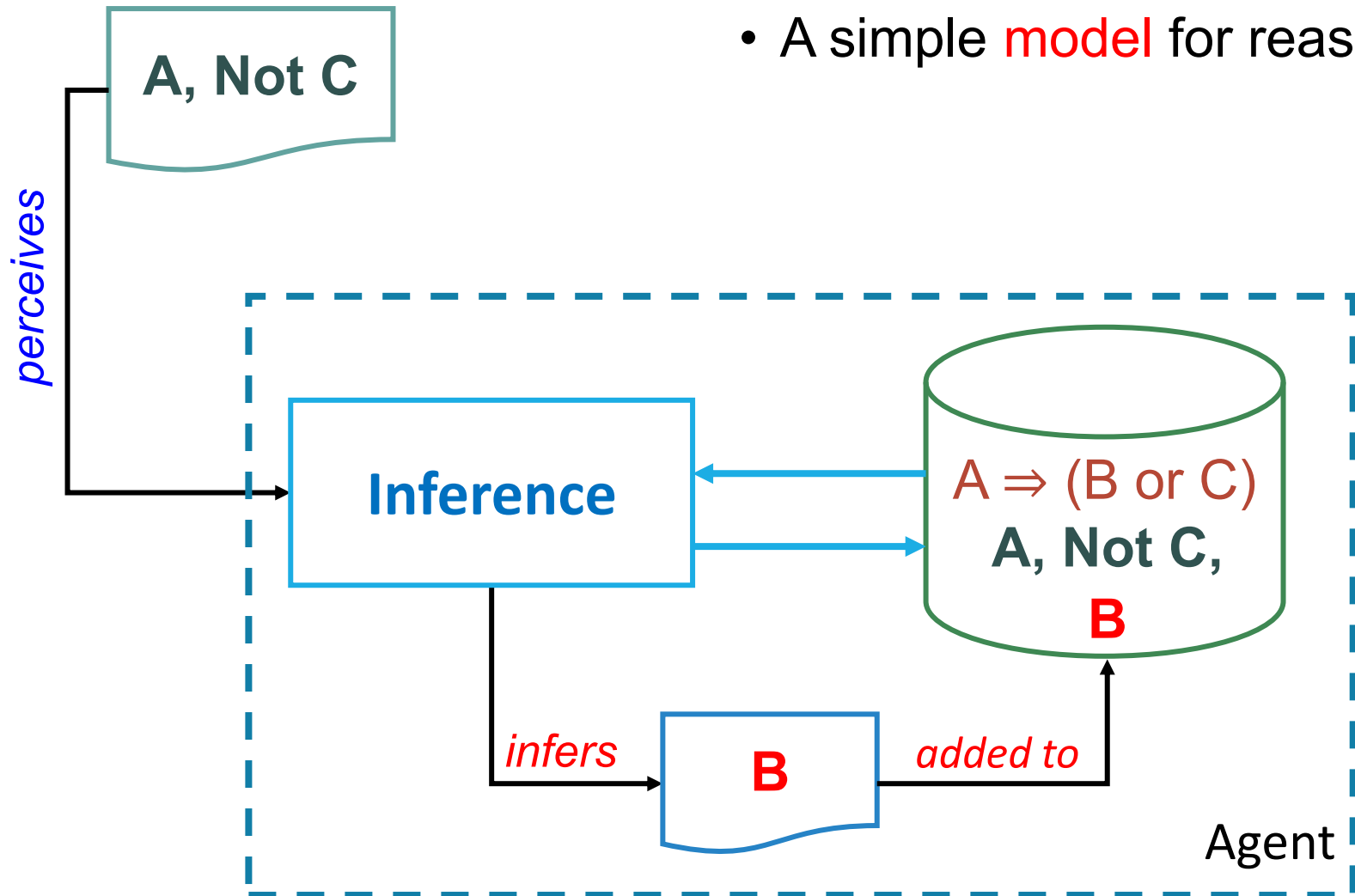
Offline and online decomposition of an agent



- **Knowledge base (KB):** A set of sentences or facts
 - Each sentence represents some assertion about the world.
 - **Axiom** = sentence that is not derived from other sentences
- **Inference:** Derive (infer) new sentences from old ones
 - Add new sentences to the KB and query what is known.

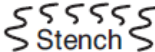
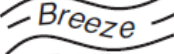
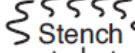
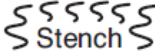
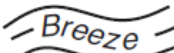
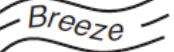
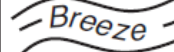
Model for reasoning: An example

- A simple **model** for reasoning



The Wumpus world



4	 Stench		 Breeze	PIT
3		  Stench  Gold	PIT	 Breeze
2	 Stench		 Breeze	
1	 START	 Breeze	PIT	 Breeze
	1	2	3	4

PEAS Description

- Environment

- 4×4 grid of rooms, agent starts in the square [1,1], facing to the right
- The locations of Gold and Wumpus are random
- Each square can be a pit, with probability 0.2

- Performance measure

- +1000 for climbing out of the cave with gold, -1000 for death
- -1 per step, -10 for using the arrow
- The game ends when agent dies or climbs out of the cave

- Actuators: *Forward, TurnLeft/TurnRight by 90°, Grab, Shoot, Climb*

- Sensors: *Stench, Breeze, Glitter, Bump, Scream*

- Percept: *[Stench, Breeze, None, None, None]*

Exploring a Wumpus world

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

Exploring a Wumpus world

A = Agent
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S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

Exploring a Wumpus world

A = Agent
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V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

Exploring a Wumpus world

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

An agent in the Wumpus world

- A Wumpus-world agent using propositional logic will have a KB of **64** distinct **proposition symbols**, **155** sentences.

$$\neg P_{1,1}$$

$$\neg W_{1,1}$$

$$B_{x,y} \Leftrightarrow (P_{x,y+1} \vee P_{x,y-1} \vee P_{x+1,y} \vee P_{x-1,y})$$

$$S_{x,y} \Leftrightarrow (W_{x,y+1} \vee W_{x,y-1} \vee W_{x+1,y} \vee W_{x-1,y})$$

$$W_{1,1} \vee W_{1,2} \vee \dots \vee W_{4,4}$$

$$\neg W_{1,1} \vee \neg W_{1,2}$$

$$\neg W_{1,1} \vee \neg W_{1,3}$$

...

Inference using Propositional logic

Propositional logic: Syntax

$$\text{Sentence} \rightarrow \text{AtomicSentence} \mid \text{ComplexSentence}$$
$$\text{AtomicSentence} \rightarrow \text{True} \mid \text{False} \mid P \mid Q \mid R \mid \dots$$
$$\text{ComplexSentence} \rightarrow (\text{Sentence})$$
$$\mid \neg \text{Sentence}$$
$$\mid \text{Sentence} \wedge \text{Sentence}$$
$$\mid \text{Sentence} \vee \text{Sentence}$$
$$\mid \text{Sentence} \Rightarrow \text{Sentence}$$
$$\mid \text{Sentence} \Leftrightarrow \text{Sentence}$$

OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Literal: atomic sentence (P) or negated atomic sentence ($\neg P$)

Logics in general

- **Models** are mathematical abstractions, each of which has a fixed truth value for every relevant sentence.
 - E.g., all possible assignments of real numbers to x and y for the sentence $x + y = 4$.
- If a sentence α is true in model m , m satisfies α or m is a model of α .
- The set of all models of α is $M(\alpha)$.

Propositional logic: Semantics

- Each model specifies true/false for each proposition symbol.
 - E.g., $m_1 = \{P_{1,2} = \text{false}, P_{2,2} = \text{false}, P_{3,1} = \text{true}\}$, 8 possible models
- Rules for evaluating truth with respect to a model m

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

- Simple recursive process evaluates an arbitrary sentence.
 - E.g., $\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$

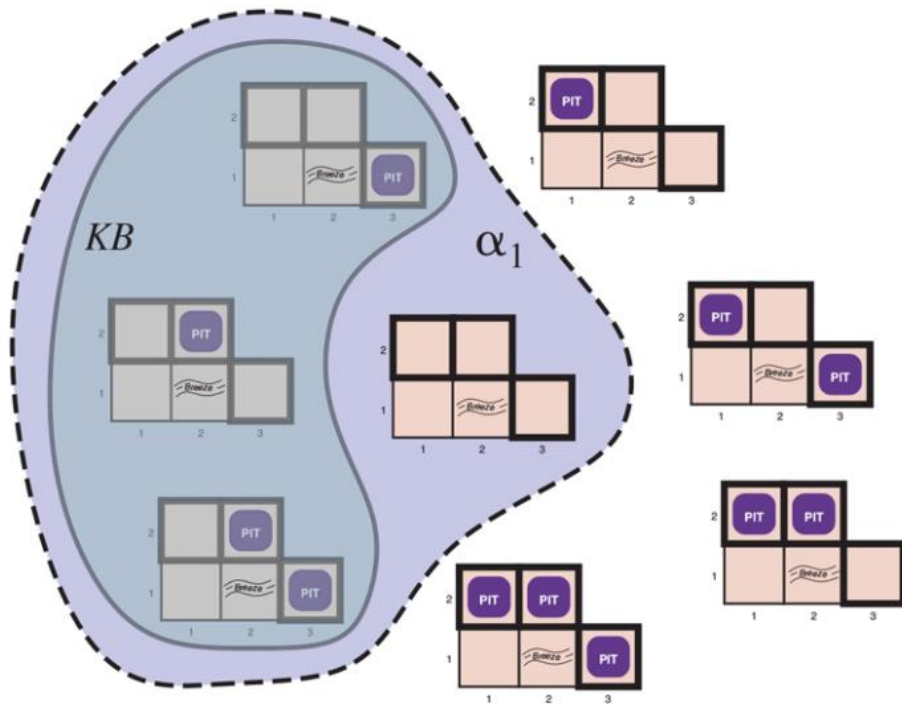
Entailment in logic

- **Entailment** is a relationship between sentences (i.e., **syntax**) that is based on **semantics**.
- **Idea:** A sentence **follows logically** from another sentence
- α entails β , $\alpha \models \beta$, if and only if, in every model in which α is true, β is also true, i.e., $M(\alpha) \subseteq M(\beta)$.
 - E.g., $x = 0$ entails $xy = 0$, while $xy = 0$ is also true when $y = 0$.

Entailment in logic: Wumpus world

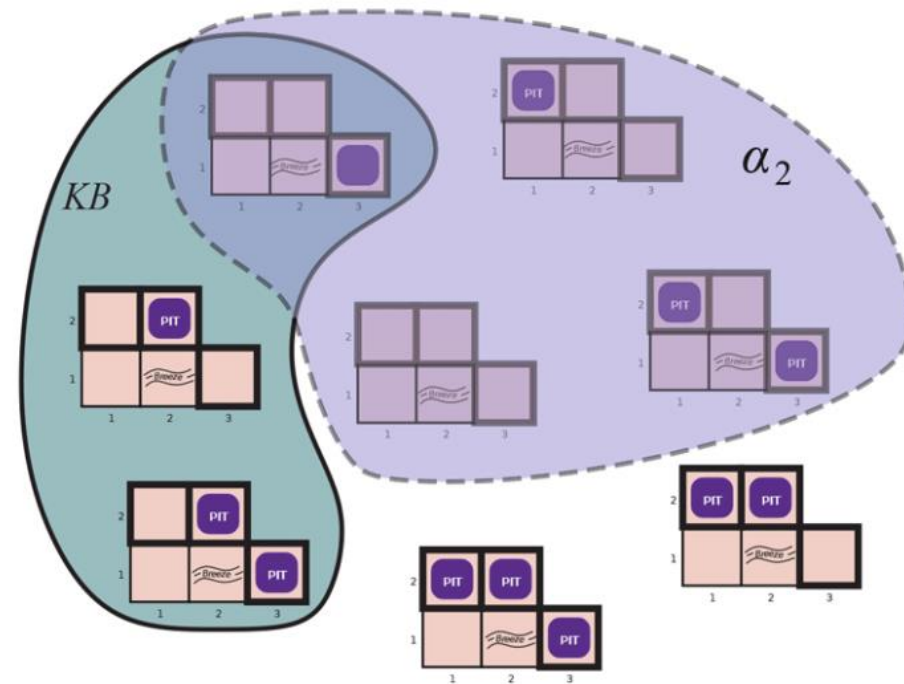
- Consider two possible conclusions α_1 and α_2

“There is no pit in [1,2].”



$KB \models \alpha_1$

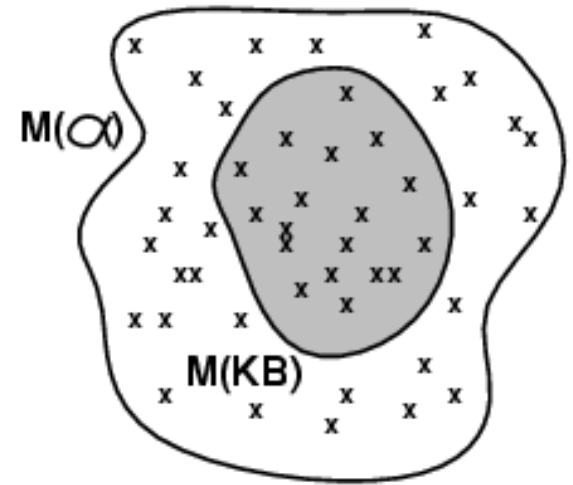
“There is no pit in [2,2].”



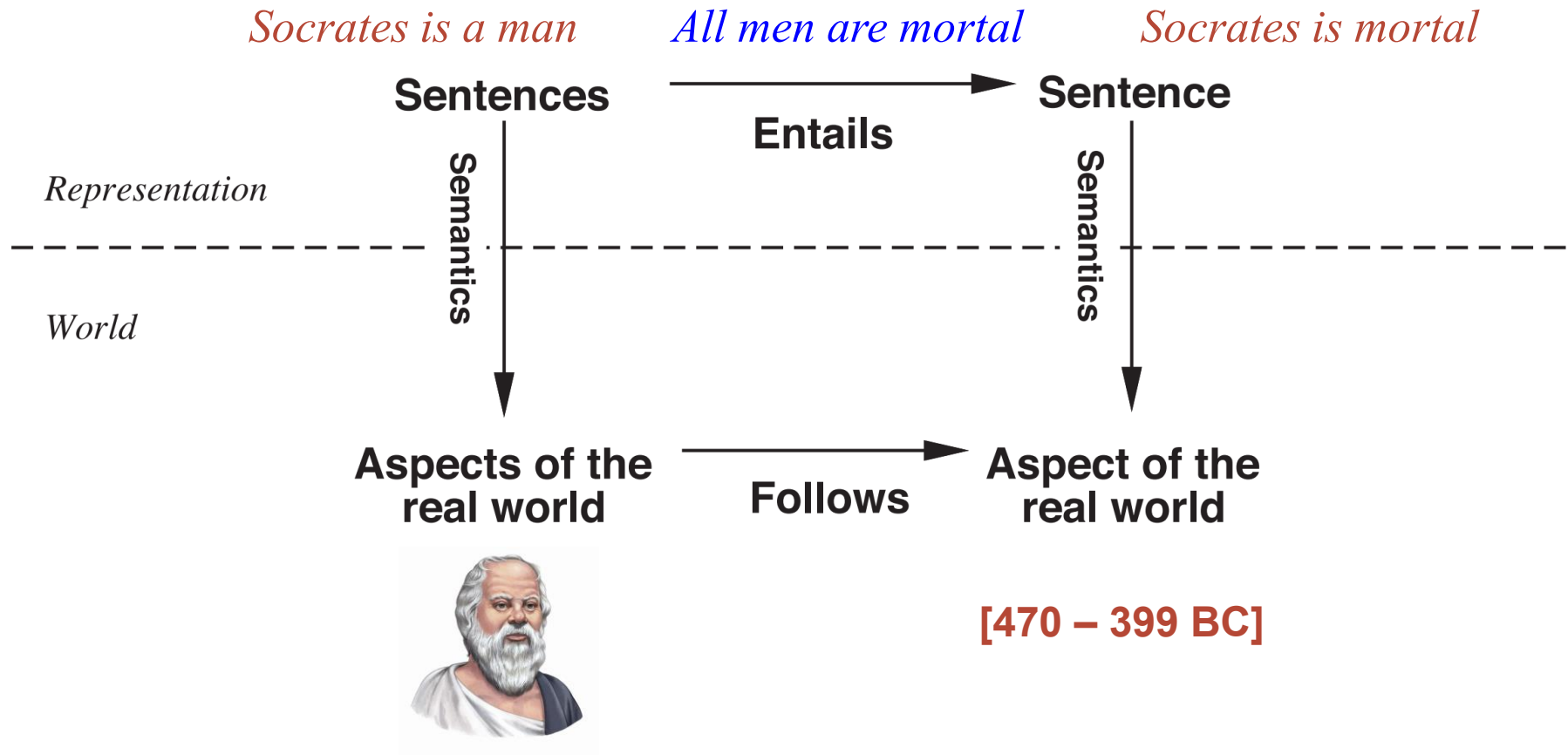
$KB \not\models \alpha_2$

Logical inference

- Let $KB \models_i \alpha$ mean α can be derived from KB by procedure i .
- Procedure i is sound if whenever $KB \models_i \alpha$, it is also true that $KB \models \alpha$
- Procedure i is complete if whenever $KB \models \alpha$, it is also true that $KB \models_i \alpha$
- That is, the procedure will answer any question whose answer follows from what is known by the KB .



World and representation



Logical reasoning should ensure that the new configurations represent aspects of the world that actually follow from the aspects that the old configurations represent.

A simple knowledge base

- Symbols for each position $[i, j]$
 - $P_{i,j}$: there is a pit in $[i, j]$
 - $W_{i,j}$: there is a Wumpus in $[i, j]$
 - $B_{i,j}$: there is a breeze in $[i, j]$
 - $S_{i,j}$: there is a stench in $[i, j]$
- Sentences in Wumpus world's KB

$$R_1: \neg P_{1,1}$$

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4: \neg B_{1,1}$$

$$R_5: B_{2,1}$$

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

A simple inference procedure

- **Given:** a set of sentences, KB , and sentence α
- **Goal:** answer $KB \models \alpha?$ = “Does KB semantically entail α ?”
 - In all interpretations in which KB 's sentences are true, is α also true?
 - E.g., in the Wumpus world, $KB \models P_{1,2}?$ = “Is there a pit in [1,2]?”

Model-checking approach (Inference by enumeration)

Inference rules

Conversion to the inverse SAT problem (Resolution refutation)

Model-checking approach

- Check if α is true in every model in which KB is true.
 - E.g., the Wumpus's KB has 7 symbols $\rightarrow 2^7 = 128$ models
- Draw a truth table for checking

No pit in [1,2]

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	false	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	true	true	true	true	true	true	<u>true</u>
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Inference by (depth-first) enumeration

function TT-ENTAILS?(KB, α) **returns** *true or false*

inputs: KB , the knowledge base, a sentence in propositional logic

α , the query, a sentence in propositional logic

$symbols \leftarrow$ a list of the proposition symbols in KB and α

return TT-CHECK-ALL($KB, \alpha, symbols, \{ \}$)

function TT-CHECK-ALL($KB, \alpha, symbols, model$) **returns** *true or false*

if EMPTY?($symbols$) **then**

if PL-TRUE?($KB, model$) **then return** PL-TRUE?($\alpha, model$)

else return *true* // when KB is false, always return true

else do

$P \leftarrow$ FIRST($symbols$)

$rest \leftarrow$ REST($symbols$)

return (TT-CHECK-ALL($KB, \alpha, rest, model \cup \{P = true\}$)

and TT-CHECK-ALL($KB, \alpha, rest, model \cup \{P = false\}$))

sound and complete

Time complexity $O(2^n)$, space complexity $O(n)$

Quiz 01: Model-checking approach

- Given a KB containing the following rules and facts
 - R_1 : IF hot AND smoky THEN fire
 - R_2 : IF alarm_beeps THEN smoky
 - R_3 : IF fire THEN sprinklers_on
 - F_1 : alarm_beeps
 - F_2 : hot
- Represent the KB in propositional logic with given symbols
 - H = hot, S = smoky, F = fire, A = alarms_beeps, R = sprinklers_on
- Answer the question “Sprinklers_on?” by using the model-checking approach.

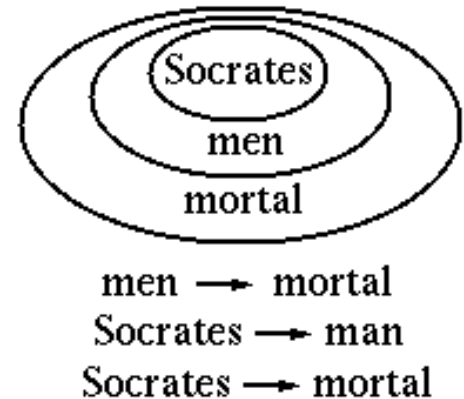


Propositional theorem proving

- *Proof by Resolution*
 - *Forward and Backward Chaining*
-

Inference rules approach

- **Theorem proving:** Apply **rules of inference directly** to the sentences in KB to construct a **proof of the desired sentence without consulting models**.
- More efficient than model checking when the number of models is large, yet the length of the proof is short
- **Proof by inference rules** is **sound** but **not complete**
 - If the rules are inadequate, then the goal is not reachable.



Logical equivalence

- Two sentences, α and β , are **logically equivalent** if they are true in the same set of models.
- Alternatively, α and β are equivalent if and only if each of them entails the other.

$$\alpha \equiv \beta \text{ iff } \alpha \models \beta \text{ and } \beta \models \alpha$$

Logical equivalence

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{De Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{De Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Validity and Satisfiability

- A sentence is **valid** if it is true in all models.
 - E.g., $P \vee \neg P$, $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$
- Valid sentences are also known as **tautologies**.
- Validity is connected to inference via the **Deduction Theorem**

$\alpha \models \beta$ iff $\alpha \Rightarrow \beta$ is valid

- A sentence is **satisfiable** if it is true in **some** model.
 - E.g., $P \vee Q$, P
- A sentence is **unsatisfiable** if it is true in **no** models.
 - E.g., $P \wedge \neg P$

Satisfiability

- **Satisfiability** is connected to inference via the following
$$\alpha \models \beta \text{ iff } \alpha \wedge \neg\beta \text{ is unsatisfiable}$$

→ **Proof by refutation** or **proof by contradiction**
- **Refutation** gives us a practical and mechanical procedure for automated reasoning.
- **SAT problem:** Determine the satisfiability of sentences in propositional logic (NP-complete)
 - This problem shares some ideas with CSPs, where the constraints are satisfiable by some assignment.

Quiz 02: Validity and Satisfiability

- Check the validity and satisfiability of the below sentences using the truth table

1. $A \vee B \Rightarrow A \wedge C$

2. $A \wedge B \Rightarrow A \vee C$

3. $(A \vee B) \wedge (\neg B \vee C) \Rightarrow A \vee C$

4. $(A \vee \neg B) \Rightarrow A \wedge B$

Inference and Proofs

- **Proof:** A chain of conclusions leads to the desired goal
- Example sound rules of inference

$$\frac{\alpha \Rightarrow \beta \quad \alpha}{\therefore \beta}$$

Modus Ponens

$$\frac{\alpha \Rightarrow \beta \quad \neg \beta}{\therefore \neg \alpha}$$

Modus Tollens

$$\frac{\alpha \quad \beta}{\therefore \alpha \wedge \beta}$$

AND-Introduction

$$\frac{\alpha \wedge \beta}{\therefore \alpha}$$

AND-Elimination

Inference rules: An example

KB	No.	Sentences	Explanation
$P \wedge Q$	1	$P \wedge Q$	From KB
$P \Rightarrow R$	2	$P \Rightarrow R$	From KB
$Q \wedge R \Rightarrow S$	3	$Q \wedge R \Rightarrow S$	From KB
$S?$	4	P	1 And-Elim
	5	R	4,2 Modus Ponens
	6	Q	1 And-Elim
	7	$Q \wedge R$	5,6 And-Intro
	8	S	3,7 Modus Ponens

Inference rules in Wumpus world

$$R_1: \neg P_{1,1}$$

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4: \neg B_{1,1}$$

$$R_5: B_{2,1}$$

Proof: $\neg P_{1,2}$

- Bi-conditional elimination to $R_2: R_6: (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
- And-Elimination to $R_6: R_7: (P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}$
- Logical equivalence for contrapositives: $R_8: \neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})$
- Modus Ponens with R_8 and the percept $R_4: R_9: \neg(P_{1,2} \vee P_{2,1})$
- De Morgan's rule: $R_{10}: \neg P_{1,2} \wedge \neg P_{2,1}$
- And-Elimination again to R_{10} to obtain $\neg P_{1,2}$

Proving by search

- Search algorithms can be applied to find a sequence of steps that constitutes a proof.
 - INITIAL STATE: the initial knowledge base
 - ACTIONS: apply all inference rules to all the sentences that match the top half of the inference rule
 - RESULT: add the sentence in the bottom half of the inference rule
 - GOAL: a state that contains the sentence need to be proved
- The proof can ignore irrelevant propositions, no matter how many of them there are → **more efficient**
 - E.g., in the Wumpus world, $B_{2,1}$, $P_{1,1}$, $P_{2,2}$ and $P_{3,1}$ are not mentioned.

Monotonicity

- The set of entailed sentences only increases as information is added to the knowledge base.

$$\textit{if } KB \models \alpha \textit{ then } KB \wedge \beta \models \alpha$$

- Additional conclusions can be drawn without invalidating any conclusion α already inferred.

Proof by Resolution

- **Resolution** is a **single inference rule** that yields a **complete inference algorithm** when coupled with any complete search algorithm.

- Unit resolution inference rule

$$\frac{l_1 \vee \cdots \vee l_k}{m}$$

$$l_1 \vee \cdots \vee l_{i-1} \vee l_{i+1} \vee \cdots \vee l_k$$

where l_i and m are **complementary literals**

- **Full resolution rule**

$$\frac{l_1 \vee \cdots \vee l_k \quad m_1 \vee \cdots \vee m_n}{l_1 \vee \cdots \vee l_{i-1} \vee l_{i+1} \vee \cdots \vee l_k \vee m_1 \vee \cdots \vee m_{j-1} \vee m_{j+1} \vee \cdots \vee m_n}$$

where l_i and m_j are complementary literals

Inference rules in Wumpus world

$$R_1: \neg P_{1,1}$$

$$R_2: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R_3: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

$$R_4: \neg B_{1,1}$$

$$R_5: B_{2,1}$$

$$R_6: \left(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1}) \right) \wedge \left((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1} \right)$$

$$R_7: \neg P_{1,2} \wedge \neg P_{2,1} \Rightarrow B_{1,1}$$

$$R_8: \neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})$$

$$R_9: \neg(P_{1,2} \vee P_{2,1})$$

$$R_{10}: \neg P_{1,2} \wedge \neg P_{2,1}$$

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2 P?	3,2	4,2
OK			
1,1 V OK	2,1 A B OK	3,1 P?	4,1

Inference rules in Wumpus world

$R_1: \neg P_{1,1}$

...

$R_{11}: \neg B_{1,2}$

$R_{12}: B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$

$R_{13}: \neg P_{2,2}$

$R_{14}: \neg P_{1,3}$

$R_{15}: P_{1,1} \vee P_{2,2} \vee P_{3,1}$

$R_{16}: P_{1,1} \vee P_{3,1}$

$R_{17}: P_{3,1}$

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

$\neg P_{2,2}$ resolves with $P_{2,2}$

$\neg P_{1,1}$ resolves with $P_{1,1}$

Proof by Resolution

- **Factoring:** the resulting clause should contain only one copy of each literal.
 - E.g., resolving $(A \vee B)$ with $(A \vee \neg B)$ obtains $(A \vee A) \rightarrow$ reduced to A
- For *any pair of sentences*, α and β , *in propositional logic*, a resolution-based theorem prover can decide whether $\alpha \models \beta$.

Conjunctive Normal Form (CNF)

- Resolution applies only to clauses, i.e., disjunctions of literals
→ Convert all sentences in KB into clauses (CNF form)
- For example, convert $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$ into CNF
$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$$

→ A conjunction of 3 clauses

Conversion to CNF

1. Eliminate $\Leftrightarrow$: $\alpha \Leftrightarrow \beta \equiv (\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$
2. Eliminate $\Rightarrow$: $\alpha \Rightarrow \beta \equiv \neg\alpha \vee \beta$
3. The operator $\neg$ appears only in literals: “move $\neg$ inwards”
 - $\neg\neg\alpha \equiv \alpha$ (double-negation elimination)
 - $\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$ (De Morgan)
 - $\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$ (De Morgan)
4. Apply the distributivity law to distribute $\vee$ over $\wedge$
 - $(\alpha \wedge \beta) \vee \gamma \equiv (\alpha \vee \gamma) \wedge (\beta \vee \gamma)$

Quiz 03: Conversion to CNF

- Convert the following sentences into CNF

1. $(A \wedge B) \Rightarrow (C \Rightarrow D)$

2. $P \vee Q \Leftrightarrow R \wedge \neg Q \Rightarrow P$

Resolution refutation algorithm

- Proving $KB \models \alpha$ by showing that $KB \wedge \neg\alpha$ is **unsatisfiable**.

function PL-RESOLUTION(KB, α) **returns** *true* or *false*

inputs: KB , the knowledge base, a sentence in propositional logic
 α , the query, a sentence in propositional logic

$clauses \leftarrow$ the set of clauses in the CNF representation of $KB \wedge \neg\alpha$

$new \leftarrow \{ \}$

loop do

for each pair of clauses C_i, C_j **in** $clauses$ **do**

$resolvents \leftarrow$ PL-RESOLVE(C_i, C_j)

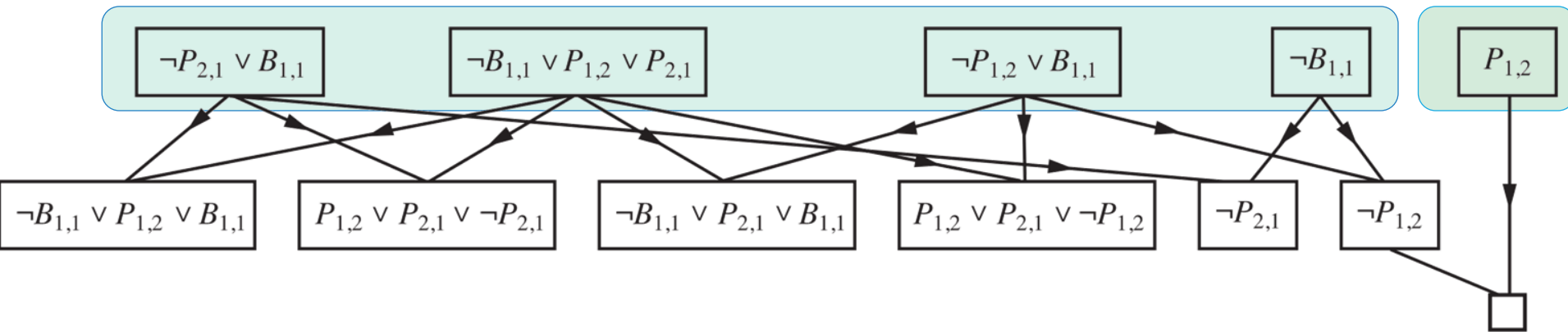
if $resolvents$ contains the empty clause **then return** *true*

$new \leftarrow new \cup resolvents$

if $new \subseteq clauses$ **then return** *false*

$clauses \leftarrow clauses \cup new$

Resolution refutation algorithm



- Many resolution steps are pointless.
- Clauses with two complementary literals can be discarded.
 - E.g., $B_{1,1} \vee \neg B_{1,1} \vee P_{2,1} \equiv \text{True} \vee P_{2,1} \equiv \text{True}$

Quiz 04: Resolution refutation

- Given the following hypotheses
 - If it rains, Joe brings his umbrella.
 - If Joe brings his umbrella, Joe does not get wet.
 - If it does not rain, Joe does not get wet.
- Prove that Joe does not get wet.

Quiz 04: Resolution refutation

- The KB contains facts and hypotheses

KB

$$R \Rightarrow U$$

$$U \Rightarrow \neg W$$

$$\neg R \Rightarrow \neg W$$

- Check if the sentence
 $\neg W$ is entailed by KB?

Horn clauses and Definite clauses

- **Definite clause:** a disjunction of literals of which **exactly one is positive**.
 - E.g., $\neg P \vee \neg Q \vee R$ is a definite clause, whereas $\neg P \vee Q \vee R$ is not.
- **Horn clause:** a disjunction of literals of which **at most one is positive**.
 - All definite clauses are Horn clauses
- **Goal clause:** clauses with **no positive literals**
- Horn clauses are closed under resolution
 - Resolving two Horn clauses will get back a Horn clause.

Propositional sentences and clauses

$$CNFSentence \rightarrow Clause_1 \wedge \dots \wedge Clause_n$$

$$Clause \rightarrow Literal_1 \vee \dots \vee Literal_m$$

$$Literal \rightarrow Symbol \mid \neg Symbol$$

$$Symbol \rightarrow P \mid Q \mid R \mid \dots$$

$$HornClauseForm \rightarrow DefiniteClauseForm \mid GoalClauseForm$$

$$DefiniteClauseForm \rightarrow (Symbol_1 \wedge \dots \wedge Symbol_l) \Rightarrow Symbol$$

$$GoalClauseForm \rightarrow (Symbol_1 \wedge \dots \wedge Symbol_l) \Rightarrow False$$

KB of definite clauses

- KB containing only definite clauses are interesting.
- Every definite clause can be written as an implication.
 - Premise (**body**) is a conjunction of positive literals and Conclusion (**head**) is a single positive literal (**fact**) → easier to understand
 - E.g., $\neg P \vee \neg Q \vee R \equiv (P \wedge Q) \Rightarrow R$
- Inference can be done with forward-chaining and backward-chaining algorithms
 - This type of inference is the basis for **logic programming**.
- Deciding entailment can be done in linear time.

KB: Horn clauses vs. CNF clauses

Disjunctions of literals
($l_1 \vee l_2 \vee \dots \vee l_m$)

CNF clauses

Clause 1

$\wedge$

Clause 2

$\wedge \dots \wedge$

Clause n

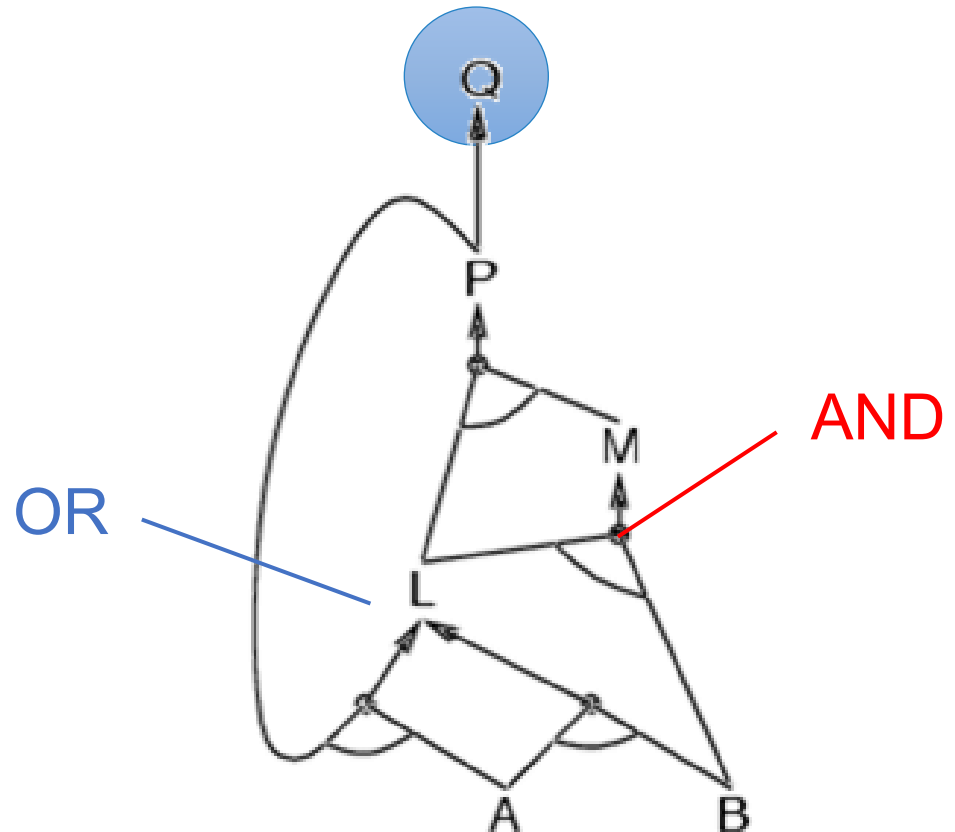
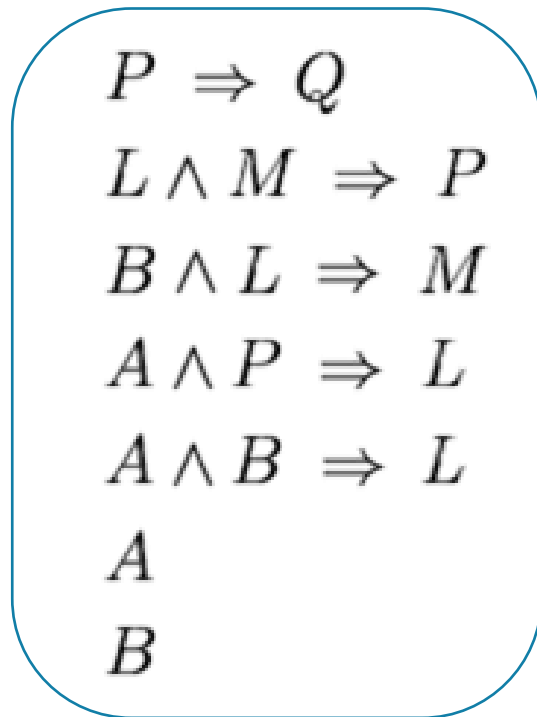
Disjunctions of literals of which **at most one is positive**
($\neg l_1 \vee \neg l_2 \vee \dots \vee l_m$)

Horn clauses

Restricted form

Forward chaining

- **Key idea:** Fire any rule whose premises are satisfied in the KB, add its conclusion to the KB, until the query is found.



Forward chaining algorithm

function PL-FC-ENTAILS?(*KB*, *q*) **returns** *true* or *false*

inputs: *KB*, the knowledge base, a set of propositional definite clauses

q, the query, a proposition symbol

count $\leftarrow$ a table, where *count*[*c*] is the number of symbols in *c*'s premise

inferred $\leftarrow$ a table, where *inferred*[*s*] is initially false for all symbols

agenda $\leftarrow$ a queue of symbols, initially symbols known to be *true* in *KB*

while *agenda* is not empty **do**

p $\leftarrow$ POP(*agenda*)

if *p* = *q* **then return** *true*

if *inferred*[*p*] = *false* **then**

inferred[*p*] $\leftarrow$ *true*

for each clause *c* in *KB* where *p* is in *c*.PREMISE **do**

decrement *count*[*c*]

if *count*[*c*] = 0 **then** add *c*.CONCLUSION to *agenda*

return *false*

Sound and complete

Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

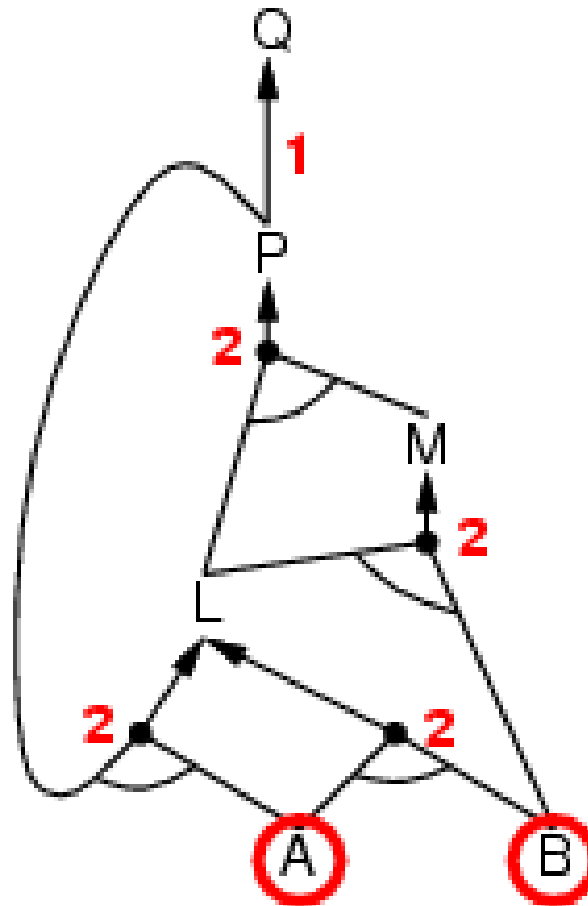
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

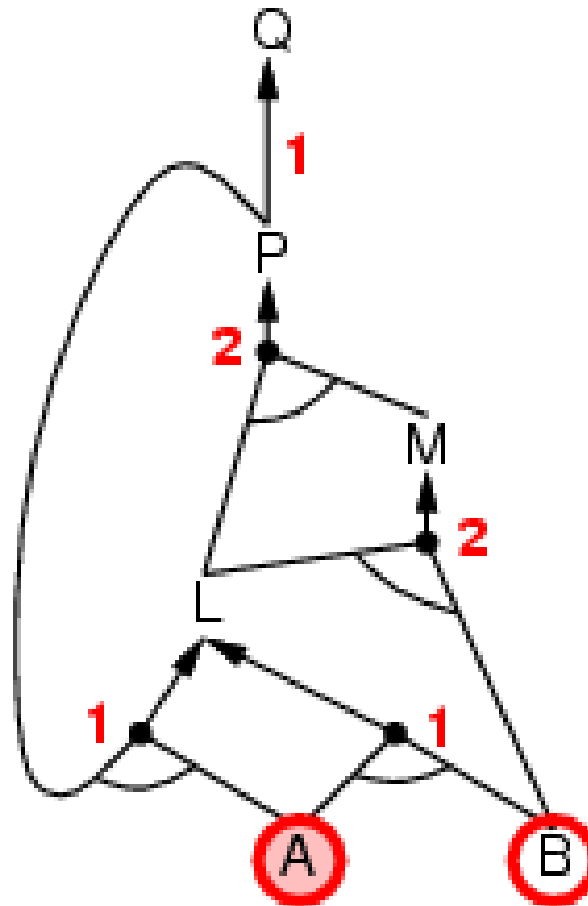
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

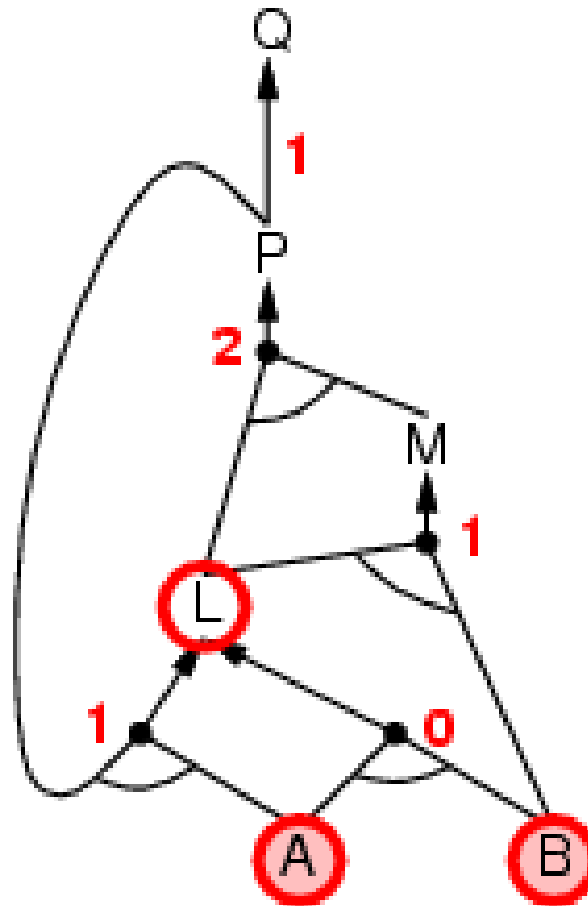
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

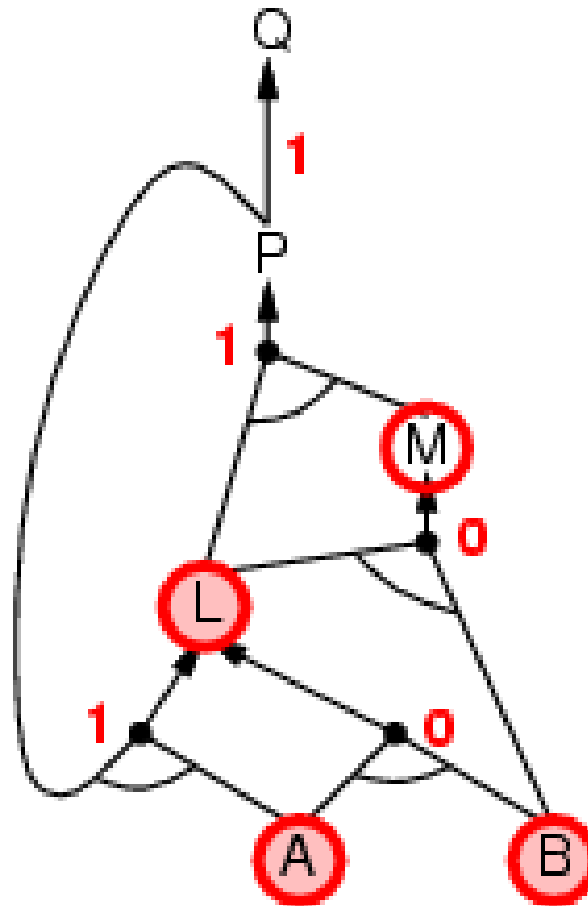
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

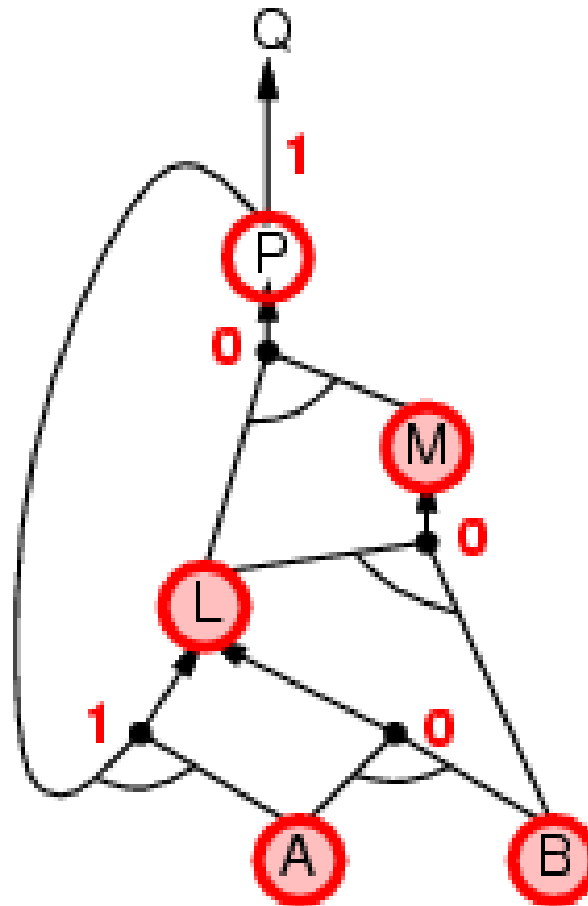
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

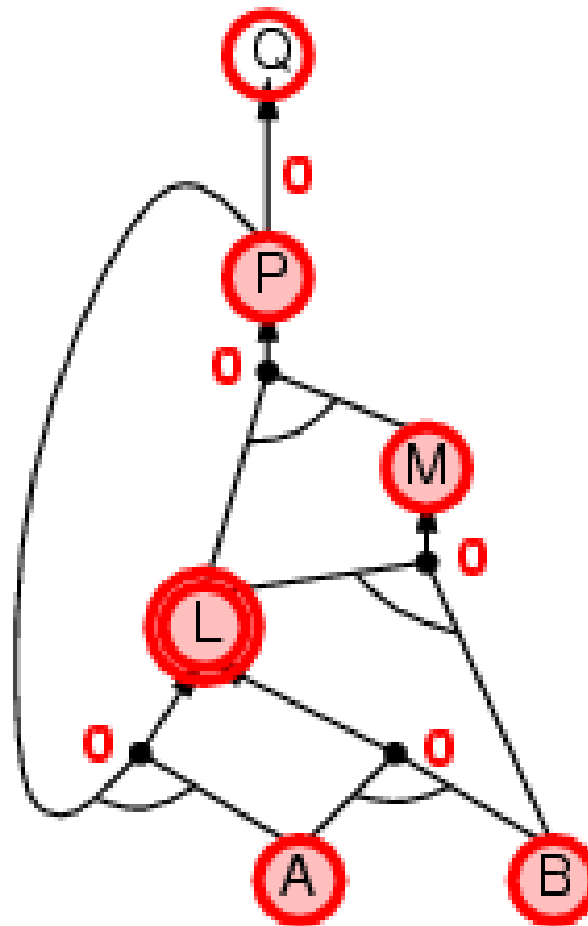
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

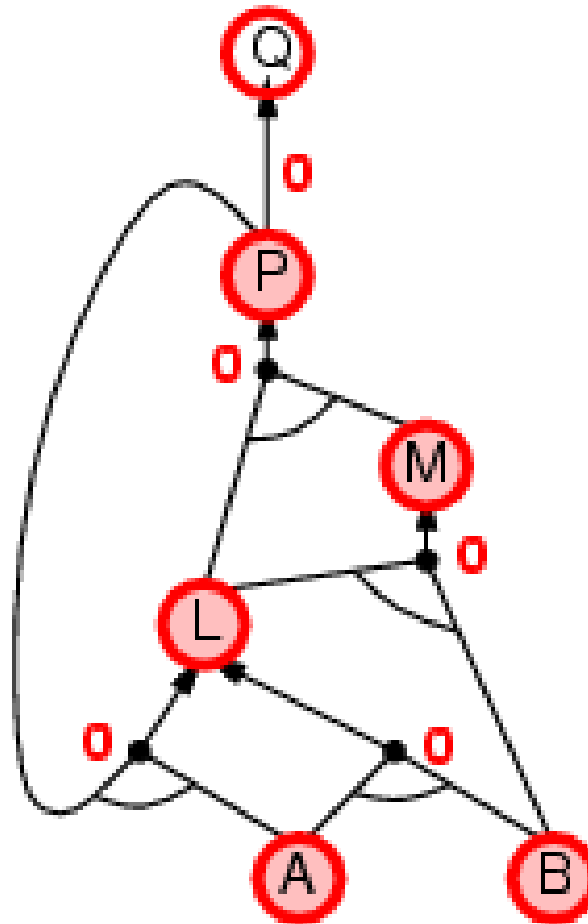
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

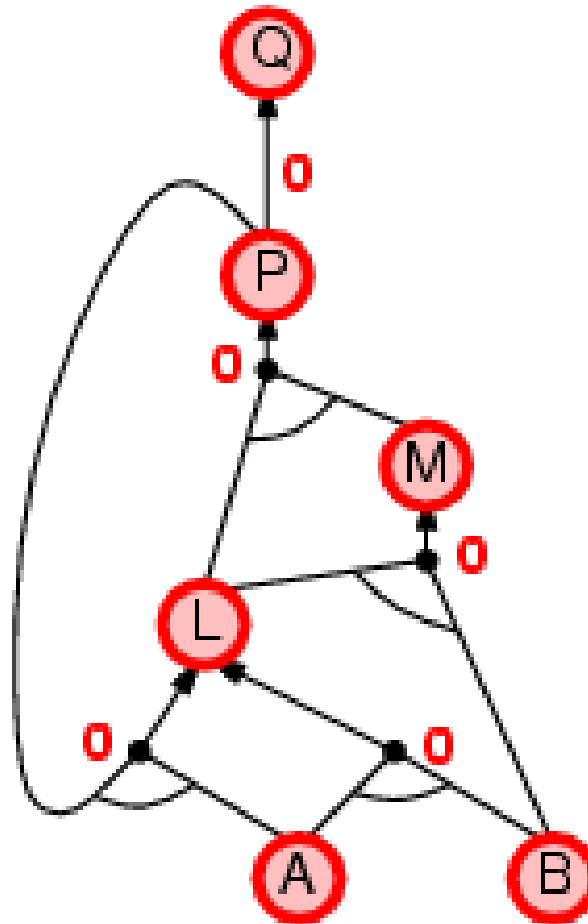
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining: Another example

KB	No.	Sentences	Explanation
$A \wedge B \Rightarrow C$	1	$A \wedge B \Rightarrow C$	From KB
$C \wedge D \Rightarrow E$	2	$C \wedge D \Rightarrow E$	From KB
$C \wedge F \Rightarrow G$	3	$C \wedge F \Rightarrow G$	From KB
A	4	A	From KB
B	5	B	From KB
D	6	D	From KB
$E?$	7	C	1, 4 and 5
	8	E	2, 6, and 7

Backward chaining

- **Key idea:** Work backwards from the query q
 - Check if q is known already, or
 - Recursively prove by BC all premises of some rule concluding q
- **Avoid loops:** A new subgoal is already on the goal stack?
- **Avoid repeated work:** A new subgoal has already been proved true, or has already failed?

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

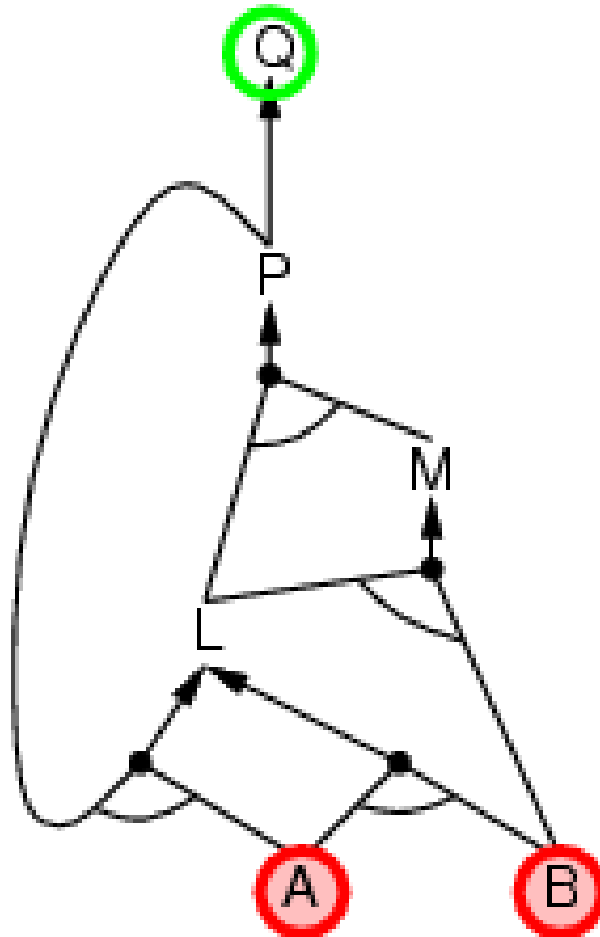
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

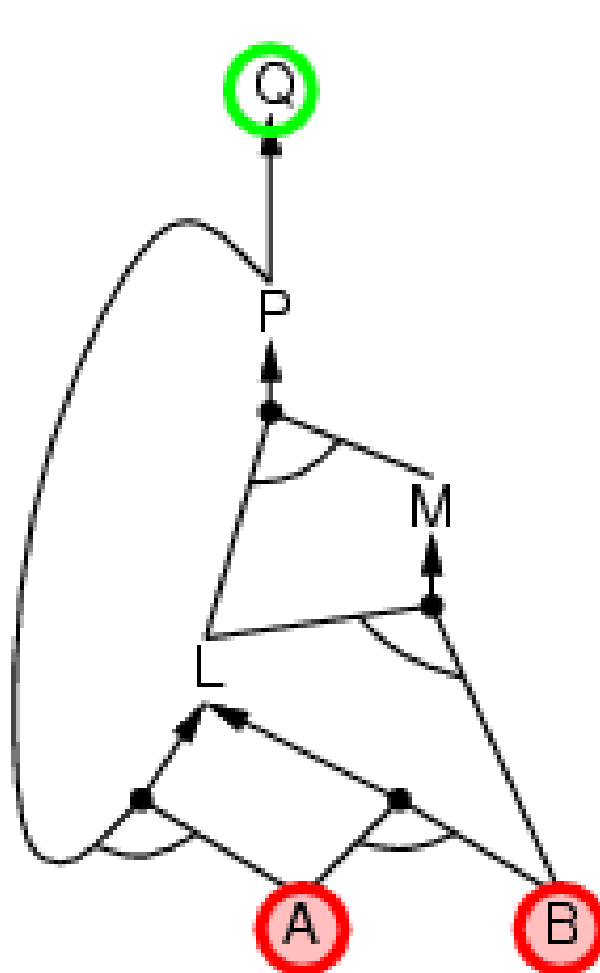
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



$Q?$
 $P?$

$P \Rightarrow Q$

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

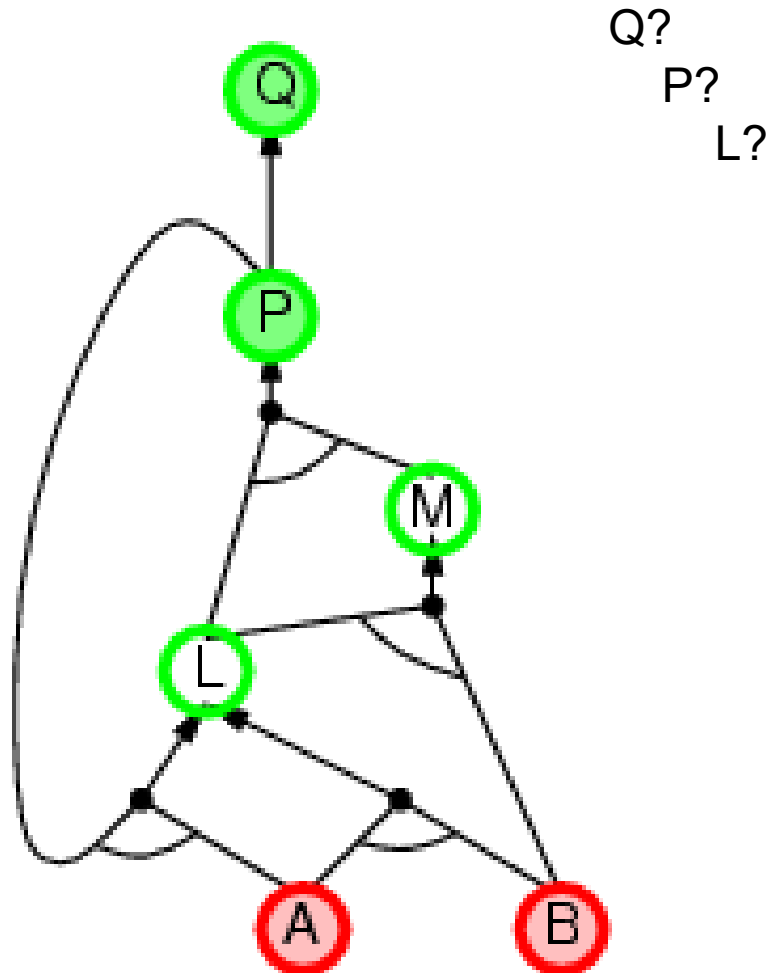
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



$$P \Rightarrow Q$$
$$L \wedge M \Rightarrow P$$

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

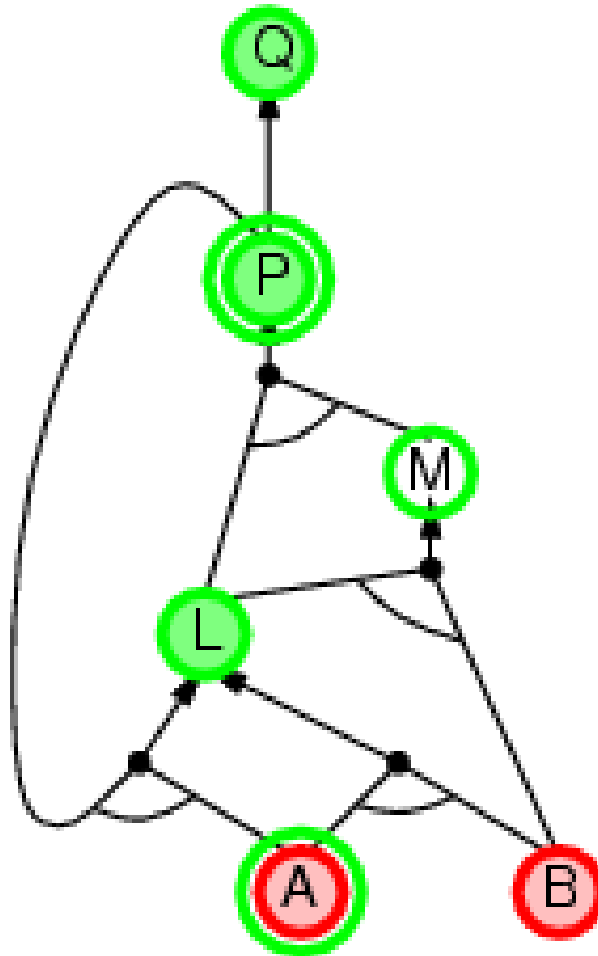
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Q?

P?

L?

A?

$P \Rightarrow Q$

$L \wedge M \Rightarrow P$

$A \wedge B \Rightarrow L$

✓

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

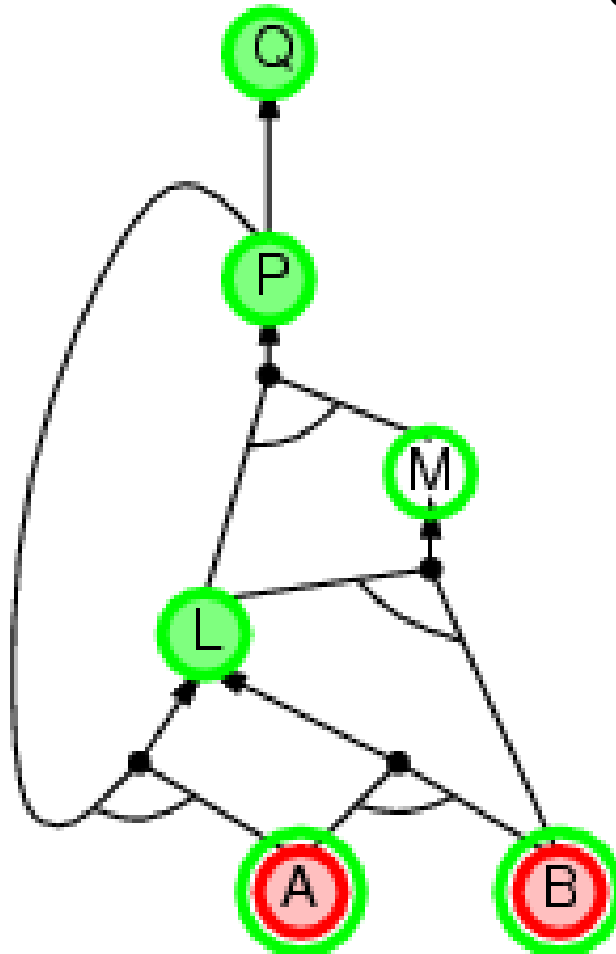
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Q?

P?

L?

A?

B?

$P \Rightarrow Q$

$L \wedge M \Rightarrow P$

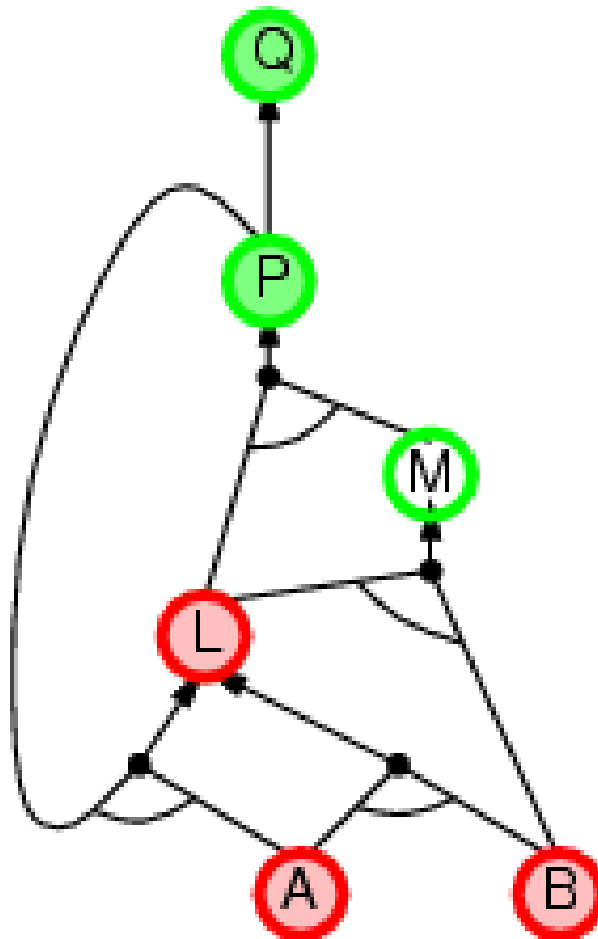
$A \wedge B \Rightarrow L$

✓

✓

Backward chaining: An example

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B

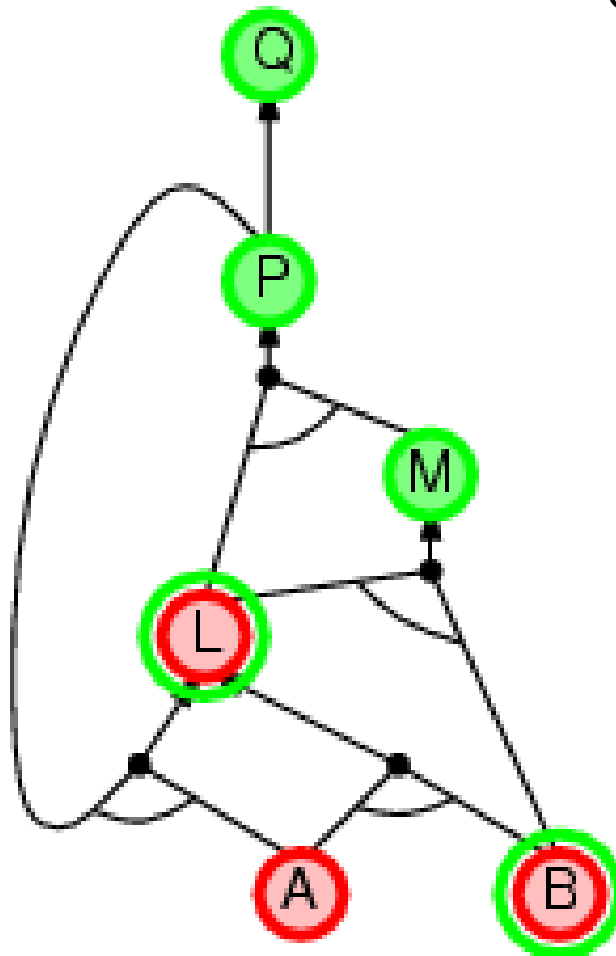


$Q?$
 $P?$
 $L?$ ✓
 $A?$
 $B?$

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 ✓
 ✓

Backward chaining: An example

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



Q?
 P?
 L? ✓
 A?
 B?
 M?
 L?
 B?

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 ✓
 ✓
 $L \wedge B \Rightarrow M$

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

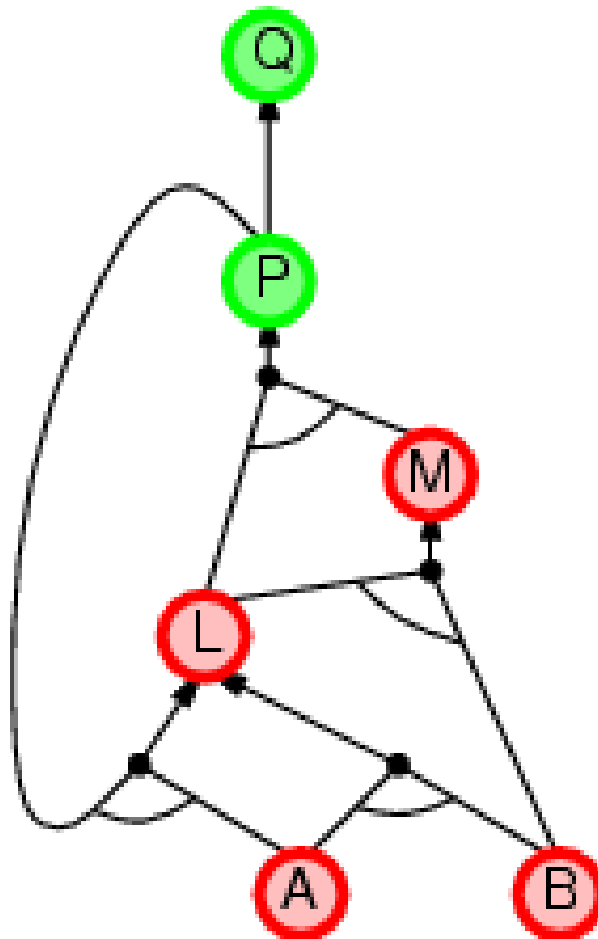
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Q?

P?

L? ✓

A?

B?

M? ✓

L?

B?

$P \Rightarrow Q$

$L \wedge M \Rightarrow P$

✓

✓

✓

✓

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

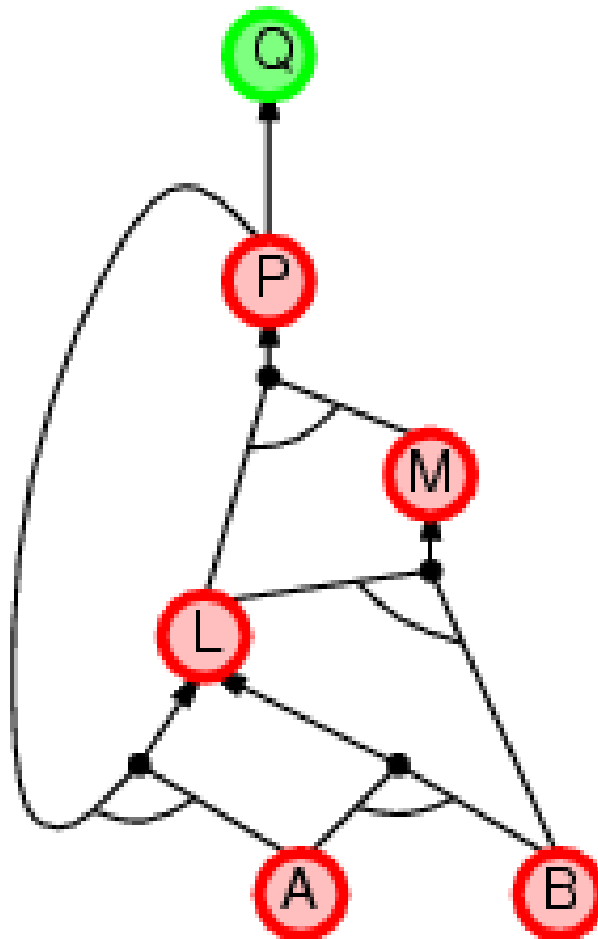
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Q?

✓

P?

✓

L? ✓

A?

✓

B?

✓

M? ✓

L?

✓

B?

✓

Backward chaining: An example

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

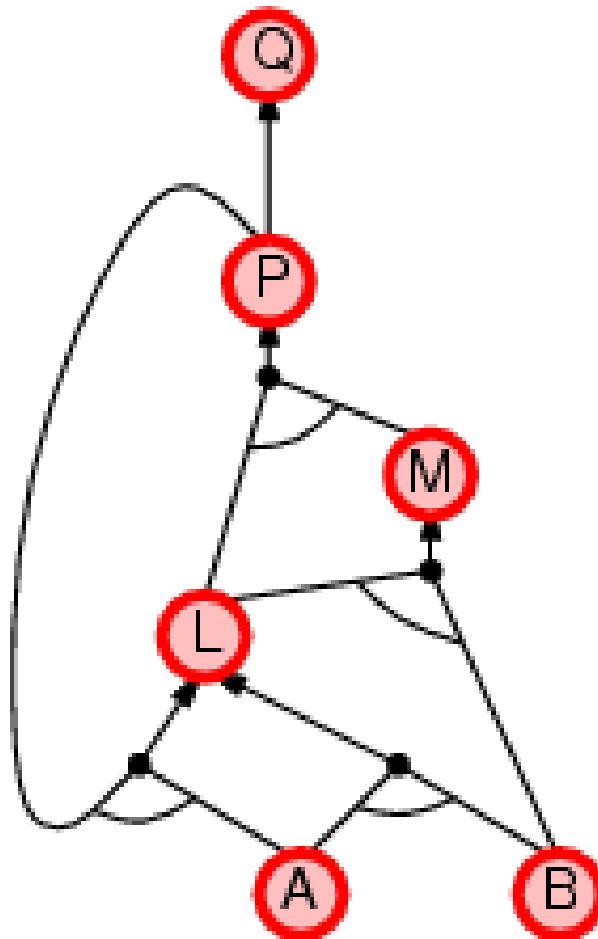
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Q?

✓

P?

✓

L? ✓

A?

✓

B?

✓

M? ✓

L?

✓

B?

✓

Backward chaining: Another example

KB

$A \wedge B \Rightarrow C$

$C \wedge D \Rightarrow E$

$C \wedge F \Rightarrow G$

A

B

D

$E?$

- $E?$

$C \wedge D \Rightarrow E$

- $C?$

$A \wedge B \Rightarrow C$

- $A?$

- $B?$

- $D?$

- A, B and D are given $\rightarrow$ All needed rules are satisfied $\rightarrow$ The goal is proven.

Forward vs. Backward chaining

- **Forward chaining:** **data-driven**, automatic, unconscious processing
 - E.g., object recognition, routine decisions
 - May do lots of work that is irrelevant to the goal
- **Backward chaining:** **goal-driven**, good for problem-solving
 - E.g., Where are my keys? How do I get into a PhD program?
 - Complexity can be **much less** than linear in size of KB

Quiz 05: Forward vs. Backward chaining

- Given a KB containing the following rules and facts
 - R_1 : IF hot AND smoky THEN fire
 - R_2 : IF alarm_beeps THEN smoky
 - R_3 : IF fire THEN sprinklers_on
 - F_1 : alarm_beeps
 - F_2 : hot
- Represent the KB in propositional logic with given symbols
 - H = hot, S = smoky, F = fire, A = alarms_beeps, R = sprinklers_on
- Answer the question “Sprinklers_on?” by using the forward chaining and backward chaining approaches



Effective model checking

- *A complete backtracking algorithm*
 - *Local search algorithms*
-

Efficient propositional inference

- There are two families of efficient algorithms for general propositional inference based on model checking
 1. Complete backtracking search algorithms
 - **DPLL** algorithm (*Davis, Putnam, Logemann, Loveland*)
 2. Incomplete local search algorithms (hill-climbing)
 - **WalkSAT** algorithm

DPLL algorithm

- Often called the **Davis-Putnam algorithm** (1960)
- Determine whether an input **propositional logic sentence (in CNF)** is **satisfiable**.
 - A recursive, depth-first enumeration of possible models.
- Improvements over truth table enumeration
 1. **Early termination**
 2. **Pure symbol heuristic**
 3. **Unit clause heuristic**

Improvements in DPLL

- **Early termination:** A clause is **true** if **any literal** is **true**, and a sentence is **false** if **any clause** is **false**.
 - Avoid examination of entire subtrees in the search space
 - E.g., $(A \vee B) \wedge (A \vee C)$ is true if A is true, regardless B and C
- **Pure symbol heuristic:** A **pure symbol** always appears with the same "sign" in all clauses.
 - E.g., $(A \vee \neg B)$, $(\neg B \vee \neg C)$, $(A \vee C)$, A and B are pure, C is impure.
 - Make a pure symbol true $\rightarrow$ Doing so never make a clause false
- **Unit clause heuristic:** there is **only one literal in the clause** and thus this literal **must be true**
 - **Unit propagation:** if the model contains $B = \text{true}$ then $(\neg B \vee \neg C)$ simplifies to a unit clause $\neg C \rightarrow C$ must be false (so that $\neg C$ is true) $\rightarrow A$ must be true (so that $A \vee C$ is true)

DPLL procedure

function DPLL-SATISFIABLE?(*s*) **returns** *true* or *false*

inputs: *s*, a sentence in propositional logic

clauses $\leftarrow$ the set of clauses in the CNF representation of *s*

symbols $\leftarrow$ a list of the proposition symbols in *s*

return DPLL(*clauses*, *symbols*, { })

function DPLL(*clauses*, *symbols*, *model*) **returns** *true* or *false*

if every clause in *clauses* is *true* in *model* **then return** *true*
if some clause in *clauses* is *false* in *model* **then return** *false* } 1. Early Termination

P, *value* $\leftarrow$ ² FIND-PURE-SYMBOL(*symbols*, *clauses*, *model*)

if *P* is non-null **then return** DPLL(*clauses*, *symbols* - *P*, *model* $\cup$ {*P*=*value*})

P, *value* $\leftarrow$ ³ FIND-UNIT-CLAUSE(*clauses*, *model*)

if *P* is non-null **then return** DPLL(*clauses*, *symbols* - *P*, *model* $\cup$ {*P*=*value*})

P $\leftarrow$ FIRST(*symbols*); *rest* $\leftarrow$ REST(*symbols*)

return DPLL(*clauses*, *rest*, *model* $\cup$ {*P*=*true*}) **or**

DPLL(*clauses*, *rest*, *model* $\cup$ {*P*=*false*})

Davis-Putnam procedure



function DP(Δ)

for φ **in** *vocabulary* (Δ) **do**

var $\Delta' \leftarrow \{ \}$;

for Φ_1 **in** Δ **for** Φ_2 **in** Δ such that $\varphi \in \Phi_1 \neg\varphi \in \Phi_2$ **do**

var $\Phi' \leftarrow \Phi_1 - \{\varphi\} \cup \Phi_2 - \{\neg\varphi\}$;

if not *tautology*(Φ') **then** $\Delta' \leftarrow \Delta' \cup \{\Phi'\}$;

$\Delta \leftarrow \Delta - \{\Phi \in \Delta \mid \varphi \in \Phi \text{ or } \neg\varphi \in \Phi\} \cup \Delta'$;

return {**if** $\{ \} \in \Delta$ **then** *unsatisfiable* **else** *satisfiable*};

function *tautology*(Φ)

$\varphi \in \Phi$ and $\neg\varphi \in \Phi$

DPLL procedure vs. DP procedure

- DP can cause a quadratic expansion every time it is applied.
 - This can easily exhaust space on large problems.
- DPLL attacks the problem by sequentially solving smaller problems.
 - Basic idea: Choose a literal. Assume true, simplify clause set, and try to show satisfiable. Repeat for the negation of the literal.
 - Good because we do not cross multiply the clause set

WalkSAT algorithm



- Incomplete, local search algorithm
- Evaluation function: **min-conflict heuristic**, to minimize the number of unsatisfied clauses
- Balance between greediness and randomness

function WALKSAT(*clauses*, *p*, *max_flips*) **returns** a satisfying model or *failure*

inputs: *clauses*, a set of clauses in propositional logic

p, the probability of choosing to do a “random walk” move, typically around 0.5

max_flips, number of flips allowed before giving up

model $\leftarrow$ a random assignment of *true/false* to the symbols in *clauses*

for *i* = 1 **to** *max_flips* **do**

if *model* satisfies *clauses* **then return** *model*

clause $\leftarrow$ a randomly selected clause from *clauses* that is false in *model*

with probability *p* flip the value in *model* of a randomly selected symbol from *clause*

else flip whichever symbol in *clause* maximizes the number of satisfied clauses

return *failure*

WalkSAT algorithm



- The algorithm returns a model → **satisfiable**
- The algorithm returns false → **unsatisfiable** **OR** more time is needed for searching
- WalkSAT cannot always detect **unsatisfiability**
 - It is most useful **when a solution is expected to exist.**
- For example,
 - An agent cannot *reliably* use WALKSAT to prove that a square is safe in the Wumpus world.
 - Instead, it can say, “I thought about it for an hour and couldn’t come up with a possible world in which the square *isn’t* safe.”

Quiz 06: DPLL and DP

- Given a KB as shown aside

KB

$$A \Rightarrow B \vee C$$

$$A \Rightarrow D$$

$$C \wedge D \Rightarrow \neg F$$

$$B \Rightarrow F$$

$$A$$

- Using either DPLL or DP to check whether KB entails each of the following sentences
 - C
 - $B \Rightarrow \neg C$

...the end.

